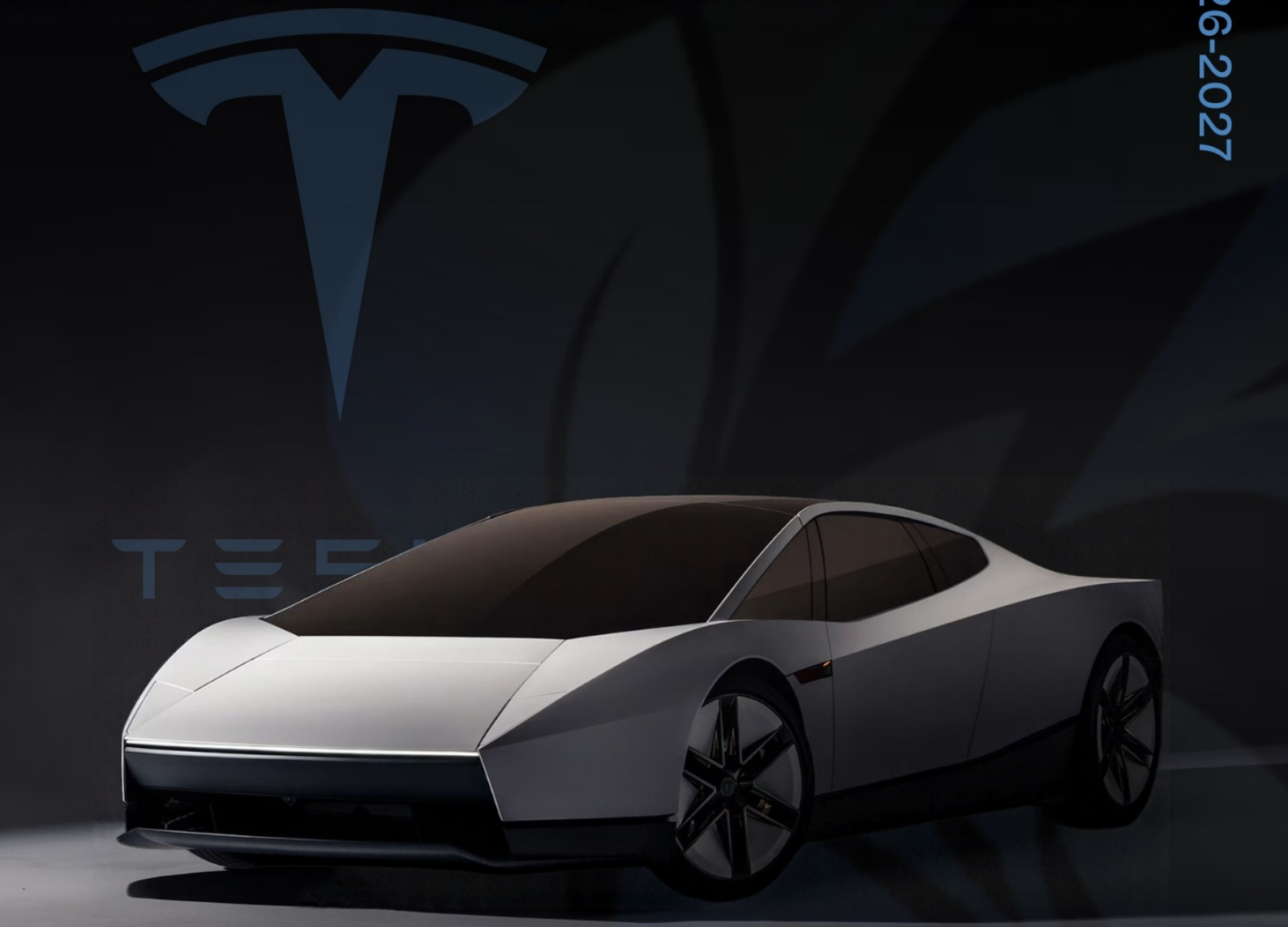
JEENISo

Business Unit 3 | 2026-2027

THE FUTURE TAKES FLIGHT

Beyond speed. Beyond roads. Beyond gravity.



Premium Edition

Engineering & Innovation.

JEENISo | National School of Engineering of Sousse

ENGINEERING MINDSET

What Truly Defines the Limit of an Engineer?

Every barrier looks permanent, right up until someone redraws it.

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Rear view. The 2026 Tesla Roadster, a car designed around a barrier rather than inside it. Image: Tesla.

Every era of industrial expansion is defined by the barriers it assumes to be permanent. In the late 19th century, prominent physicists argued that heavier-than-air powered flight was mathematically absurd because of mass-to-lift scaling laws. In the 1940s, aerodynamicists believed aircraft structures would disintegrate if they approached Mach 1, treating the sonic shockwave as a literal physical barrier. Yet in every instance, what was perceived as a fundamental law of physics was merely a failure of architectural imagination.

For student engineers at the National School of Engineers of Sousse (ENISo), developing a rigorous method to categorize challenges is the first prerequisite for innovation. Engineering limits exist in two regimes: *Thermodynamic Hard Limits* and *Architectural Soft Limits*.

Hard limits are dictated by the underlying geometry of the universe: the speed of light (c), Carnot thermodynamic efficiency, conservation of energy and quantum uncertainty. Soft limits, by contrast, are self-imposed. They stem from standardized manufacturing processes, inherited software abstraction layers, component availability, or entrenched industry heuristics.

The tragedy of modern engineering education is that students are frequently taught soft limits with the same dogmatic reverence as hard physical laws. When an engineer mistakes a soft limit for a hard limit, design halts entirely.

Consider the modern high-performance electric vehicle. For over a decade, performance gains were extracted through incremental iteration: tuning inverter switching frequencies, optimizing battery cell density, and tweaking mechanical differential lock ratios.

However, when attempting to push 0 to 60 mph acceleration below two seconds, engineers ran into the fundamental boundary of asphalt-rubber friction. No matter how many kilowatts the battery could deliver to the motors, the tires simply lost traction and spun in place.

Breaking this boundary required abandoning the implicit rule that a car must interact with the world solely through four rubber contact patches. By cross-pollinating aerospace reaction control systems with automotive design, the boundary was redrawn.

HARD LIMITS · DICTATED BY THE GEOMETRY OF THE UNIVERSE

Speed of light (c) · Carnot efficiency · Conservation of energy · Quantum uncertainty

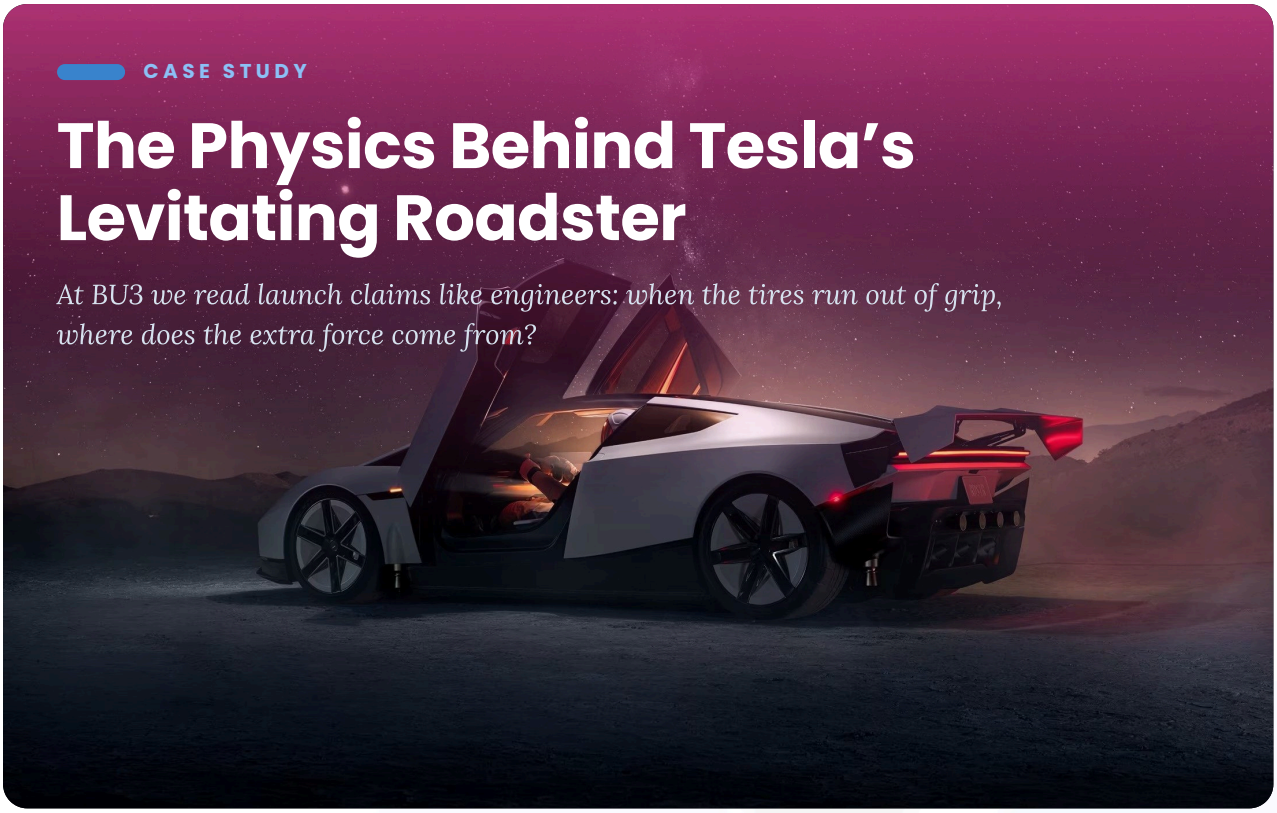
SOFT LIMITS · SELF-IMPOSED, THEREFORE NEGOTIABLE

Manufacturing standards · Software abstraction layers · Component availability · Industry heuristics

CASE STUDY

The Physics Behind Tesla's Levitating Roadster

At BU3 we read launch claims like engineers: when the tires run out of grip, where does the extra force come from?



Night run. The Roadster in its Rocket configuration, rear thruster ports visible. Image: Tesla.

Levitation, but not the magnetic kind

The word "levitating" makes us picture maglev trains or science-fiction anti-gravity. The Roadster uses neither. At its October 2026 unveiling Tesla presented three versions: Performance, Thrust and Rocket. The Thrust and Rocket trims carry high-pressure air tanks and rear nozzles based on cold-gas thrusters, a technology borrowed from spacecraft. Only the Rocket adds downward nozzles at each corner, which let it hover for a few seconds.

There is no new physics here. It is Newton's third law applied with unusual boldness: throw mass backwards, or downwards, and the car is pushed the other way. Let us see what the numbers say.

How a cold-gas thruster works

Air is stored at 10,000 psi (about 690 bar), in 200 L on the Thrust and up to 600 L on the Rocket. A valve opens, the air expands through a nozzle and leaves at high speed. The force follows the rocket equation:

$$F = (dm/dt) \cdot v_e + (p_e - p_a) \cdot A_e$$

No combustion is involved, which makes the plumbing simple: tank, valve, nozzle. The price is a low specific impulse, roughly 70 s for air against 250–450 s for chemical rockets. Cold gas gives enormous force, but only for seconds.

THE TRACTION GAP

Average acceleration implied by 0–60 mph times (26.8 m/s)

≈16.7 kN
Car weight (under 1,700 kg)

≈80 kN
Rear thrust quoted by Tesla

≈4.8×
Thrust-to-weight ratio



a = 26.8 m/s ÷ t. *Timed with roughly a one-foot rollout. Times as quoted by Tesla; g-values are our calculation.

The fan trick: suction as a second source of grip

Grip is limited by $a_{max} = \mu\,(g + F_{down}/m)$. Tires have a ceiling on μ, so the other lever is downforce. Tesla describes a multi-mode suction system on the Roadster: four fans of roughly 20 kW each that pull air from beneath the car, so the atmosphere above pushes it onto the road.

It is not a new idea. Chaparral's 2J "sucker car" raced in 1970, and Brabham's fan car won its only Formula 1 start in 1978 before being withdrawn. Tesla's display shows about 4,700 kg of downforce for a car under 1,700 kg. Independent analysts call that surprisingly large, and we agree it deserves independent testing.

TRY IT YOURSELF

If 4,700 kg of downforce acts on a sealed floor of about 4 m², what pressure difference is needed? *Answer: 4,700 × 9.81 ÷ 4 ≈ 11.5 kPa, over a tenth of an atmosphere.*



Thrust. Four rear ports fire. Debris and gas plumes reached tens of meters at the launch. Image: Tesla.



Augments. Full-width light bar, adjustable rear wing and four thruster ports. Image: Tesla.

Hovering is a budget, not a miracle

To hover, thrust must at least equal weight: $T \geq mg \approx 16.7\ kN$. Ideal expansion of air from 690 bar gives an exhaust speed of about 700 m/s, so the mass flow is $dm/dt = T / v_e \approx 24\ kg/s$.

A 200 L tank at that pressure holds roughly 110 kg of air, so hover lasts about five seconds, which matches the burst length reported for the Rocket. At the full 80 kN rear thrust the same air is gone in about one second. These are our own order-of-magnitude estimates, not Tesla data.

Hover is also a control problem

Hovering on thrust is an unstable equilibrium, like balancing a broom on a finger, and every student here knows how quickly that goes wrong. The Rocket's corner nozzles, plus two at the front for yaw, must be modulated by a fast feedback loop reading attitude sensors. This is the reaction control system of a spacecraft, transplanted into a road car.

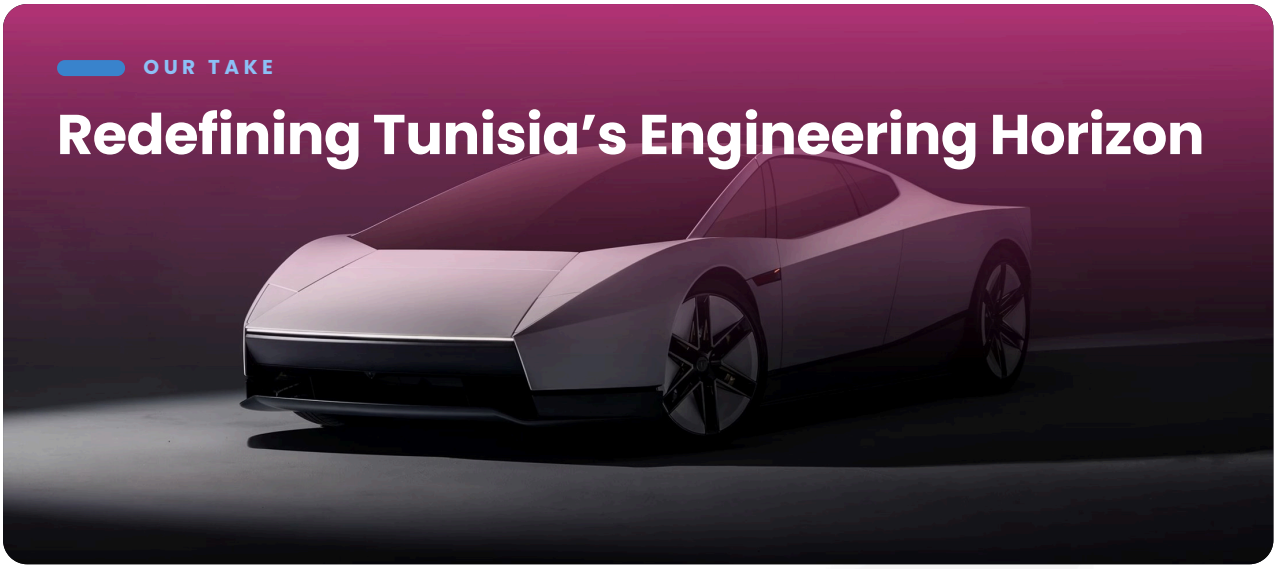
OUR VERDICT AS ENGINEERS

Real engineering, short-lived performance. For us, cold gas buys seconds of extraordinary force, which suits a launch or a demonstration far better than a commute. Note also that the range target has moved from the 600 miles promised in 2017 to "over 300", and production is reported for 2028. Claims shift; physics does not. That is the lesson we take from it.

CLAIM	WHAT IT IMPLIES	OUR CHECK
0–60 mph in 0.55 s (top spec)	≈5 g average	Grip alone caps near 1–1.5 g, so thrust and suction must do the rest.
≈80 kN of rear thrust	≈4.8× car weight	Coherent, but empties a 200 L tank in about a second.
Hover bursts of ≈5 s	≈110 kg of air at hover thrust	Consistent. Rocket is invite-only and not approved for public roads.

OUR TAKE

Redefining Tunisia's Engineering Horizon



Front quarter. The Roadster in studio light. Image: Tesla.

The ultimate takeaway of thruster-augmented vehicle engineering is not simply about breaking acceleration records. For student engineers across Junior Enterprise ENISo and nationwide, it is a powerful call to dismantle mental silos. Aerospace engineers, embedded system developers, mechanical designers and control theorists must learn to speak a unified language.

When JEENISo delivers industrial consulting projects, from automated manufacturing lines to intelligent embedded systems, our engineers must adopt the same first-principles approach: *never treat legacy process limitations as unalterable physical laws.*

As young Tunisian engineers navigating an increasingly competitive global landscape, our true competitive advantage lies in interdisciplinary synthesis. The real boundaries of engineering do not live inside software manuals or university textbooks. They exist only until someone applies principles from one discipline to conquer an unsolvable barrier in another.

01

Ask which limit is soft

Before accepting "it can't be done", name the physical law behind it. If you cannot, it is probably a habit.

02

Borrow across fields

Tires hit a ceiling; aerospace had a different answer. Look for the discipline that already solved your problem.

03

Check the numbers

Ambition earns attention, arithmetic earns trust. Estimate first, then believe, as we just did together.

> *The levitating Roadster proves that when you combine first-principles physics with audacious execution,* **even gravity becomes negotiable.**

BU3 PODCAST

فتح آفاق على البودكاست

استمعوا إلى النقاش الكامل حول هذا الموضوع في بودكاست Business Unit 3، وتابعوا الحلقة على موقعنا.

Join us for the conversation behind this article on the BU3 podcast, on our website.

Sources. Electrek, "An updated technical look at Tesla's next-gen Roadster" (15 Oct 2026); Not a Tesla App, Roadster specs and trims; Tesla launch coverage, Oct 2026. Specifications are manufacturer claims, not independently verified. Images: Tesla.



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The Next Frontier of Automotive Engineering

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